Unit 3 Guided Notes — Equations & Inequalities

Name _____ Date _____ Period _____

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

🎯 By the end of this unit, I can…

- Solve multi-step equations, including ones with variables on both sides
- Distribute and combine like terms accurately
- Identify equations with one solution, no solution, or infinitely many

Standards: NC.8.EE.7

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 One-Step Equations / Inverse Operations

📖 Words to know

Variable: a letter, like x, that stands for a number we do not know yet.

Coefficient: the number in front of a variable. In 3x, the coefficient is 3.

Constant: a number by itself, with no variable. In 3x + 5, the constant is 5.

Like terms: terms with the same variable part, like 3x and 2x, or 5 and 8.

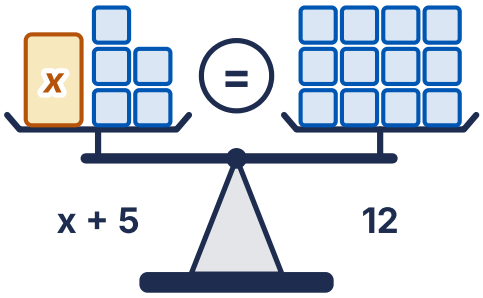
Isolate: get the variable alone on one side, like x = 4.

Solution: the number that makes the equation true.

Inequality: a math sentence with <, >, ≤ or ≥ instead of =.

📌 **THE BIG IDEA:** Use the opposite operation (the INVERSE OPERATION) to get x alone. Do the same thing to both sides. Example: x + 5 = 12 → subtract 5 from both sides → x = 7.

x tile +1 tile red = negative (−x or −1) a big block = that number



x + 5 = 12 is balanced. Keep it balanced!

Operation	Inverse	Example
Addition	_____	x + 5 = 12
Subtraction	_____	x − 3 = 8
Multiplication	_____	3x = 15
Division	_____	$\frac{x}{4} = 2$

Match the Operation to its Inverse

MATCH **Write the letter of each operation's inverse on the line.**

_____ 1. Addition

_____ 2. Subtraction

_____ 3. Multiplication

_____ 4. Division

A. Division

B. Subtraction

C. Multiplication

D. Addition

✏ Practice: One-Step Equations

1. ★ Try it Solve: x + 9 = 15

Answer: _____

2. ★★ On level Solve: $\frac{x}{3} = -4$

Answer: _____

3. ★★★ Challenge Mia solved x − 8 = **?** and got x = 3. What number is the **?**

Answer: _____

2 Two-Step Equations

📌 **TWO STEPS:** These are the last 2 of the 5 steps. Step 1 — Undo + or −. Step 2 — Undo × or ÷.

Example 1: −2x + 4 = 8

SOLVE **−2x + 4 = 8**

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by −2. The opposite is dividing, so divide BOTH sides by −2.

Answer: _____

Example 2: $\frac{x}{2} + 6 = 7$

SOLVE $\frac{x}{2} + 6 = 7$

Step 1: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 2: Get x by itself. x is DIVIDED by 2. The opposite is multiplying, so multiply BOTH sides by 2.

Answer: _____

✏ Practice: Two-Step Equations

1. ★ Try it Solve: 3x + 5 = 20

Answer: _____

2. ★★ On level Solve: −3x + 4 = 19

Answer: _____

3. ★★★ Challenge Leo solved $4x - 6 = 18$ and wrote $4x = 12$, so $x = 3$. Find his mistake. What is the correct answer?

Answer: _____

3 Distributive Property

📌 **THE RULE:** Multiply the outside number by every term inside. Watch the signs: $-2(x - 3) = -2x + 6$.

Example: $2(2x + 3) = -6$

SOLVE **$2(2x + 3) = -6$**

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 4. The opposite is dividing, so divide BOTH sides by 4.

Answer: _____

ERROR ANALYSIS A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?

✏ Practice: Distributive Property

1. ★ Try it Solve: $3(x + 2) = 21$

Answer: _____

2. ★★ On level Solve: $-2(x - 5) = 16$

Answer: _____

3. ★★★ Challenge Solve: $5(x - 1) - 2x = 13$

Answer: _____

4 Variables on Both Sides

📌 **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: $5x + 3 = 2x + 12$

SOLVE **$5x + 3 = 2x + 12$**

Step 1: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Step 2: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 3. The opposite is dividing, so divide BOTH sides by 3.

Answer: _____

✏ Practice: Variables on Both Sides

1. ★ Try it Solve: $4x + 1 = x + 10$

Answer: _____

2. ★★ On level Solve: $2x + 9 = 5x - 6$

Answer: _____

3. ★★★ Challenge Plan A costs $20 plus $3 per movie. Plan B costs $5 per movie. After how many movies do they cost the same?

Answer: _____

5 Multi-Step Equations — The 5-Step Process

📌 **THE 5 STEPS:** 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

SOLVE **$-2x + 1 = 8(x + 2) - 5x$**

Step 1: Distribute. Multiply 8 by EACH term inside the ( ).

Step 2: Combine like terms. Put like terms together (x terms with x terms, numbers with numbers): 8x − 5x = 3x.

Step 3: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Step 4: Get rid of the + 1. The opposite of + 1 is − 1, so subtract 1 from BOTH sides.

Step 5: Get x by itself. x is MULTIPLIED by −5. The opposite is dividing, so divide BOTH sides by −5.

Answer: _____

✏ Practice: Multi-Step Equations

1. ★ Try it Solve: $2(x + 3) + 4x = 30$

Answer: _____

2. ★★ On level Solve: 3(x − 2) = x + 8

Answer: _____

3. ★★★ Challenge Solve: 4(x + 1) − 2x = 3(x − 2) + 7

Answer: _____

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like 5 = 9).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like 6 = 6).

Result	Meaning	Example
Variables stay; x = number	_____	2x+1=7 → x=3
Variables cancel; FALSE remains	_____	2x+1=2x+5 → 1=5 ×
Variables cancel; TRUE remains	_____	2x+1=2x+1 → 1=1 ✓

No Solution — Solve: $3(x + 2) = 3x + 10$

SOLVE $3(x + 2) = 3x + 10$

Step 1: Distribute. Multiply 3 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Answer: _____

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

SOLVE $2(x + 4) = 2x + 8$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Answer: _____

Quick Check

MC **Solving an equation gives 7 = 7. The equation has...**

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions

✏ Practice: Special Cases

1. ★ Try it You solve an equation. The x's cancel and you get 2 = 9. How many solutions?

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions

2. ★★ On level How many solutions does $5(x - 2) = 5x - 10$ have?

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions

3. ★★ On level How many solutions does $3x + 1 = 2x + 1$ have?

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions

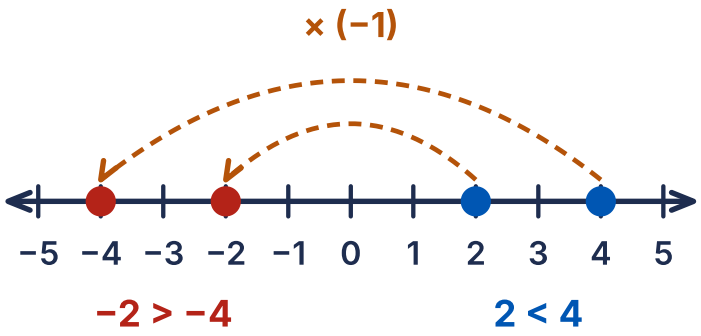
4. ★★★ Challenge What number goes in the box so $2(x + 5) = 2x + \square$ has **infinitely many** solutions?

Answer: _____

7 Linear Inequalities

Symbol	Meaning	Circle	Shade
<	_____	_____	_____
>	_____	_____	_____
≤	_____	_____	_____
≥	_____	_____	_____

⚠️ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: −2x > 6 → divide by −2 → x < −3 (sign flipped!)



The order flips, so the sign flips.

Example (no flip): $2x + 3 > 11$

SOLVE $2x + 3 > 11$

Step 1: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by 2. The opposite is dividing, so divide BOTH sides by 2.

Answer: _____

Graph the answer:



Example (FLIP NEEDED): $-3x + 4 \geq 13$

SOLVE $-3x + 4 \geq 13$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

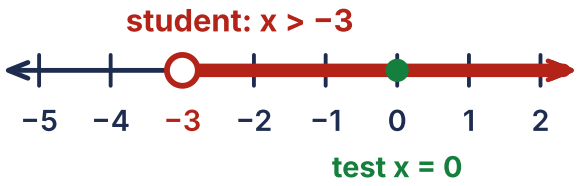
Step 2: Get x by itself. x is MULTIPLIED by −3. The opposite is dividing, so divide BOTH sides by −3. −3 is negative, so the inequality sign FLIPS.

Answer: _____

Graph the answer:



ERROR ANALYSIS To solve $-4x > 12$ a student wrote $x > -3$. What's the mistake?



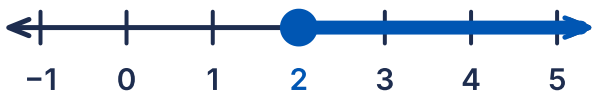
Test $x = 0$ in $-4x > 12$. Is $-4(0) > 12$ true?

✏ Practice: Linear Inequalities

1. ★ Try it Solve: $x + 6 \leq 10$.

Answer: _____

2. ★ Try it What inequality does this number line show?



Answer: _____

3. ★★ On level Solve: $-2x + 5 > 11$.

Answer: _____

4. ★★★ Challenge Dev solved $-5x \leq 20$ and got $x \leq -4$. Find his mistake. What is the correct answer?

Answer: _____

8 Word Problems — Equations & Inequalities

📌 **WORD PROBLEMS IN 4 STEPS:** 1) Let x = the thing you don't know. → 2) Write the equation. → 3) Solve it with the 5 steps. → 4) Answer in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least (≥), at most (≤), more than (>), fewer than (<)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER **Movie Theater: A movie theater charges a $6 entry fee plus $4 per ticket. If the total cost was $30, how many tickets were purchased?**

Let _____

Equation: _____

Solve (show each step):

Answer: _____

SHORT ANSWER **Notebooks: A student wants to spend no more than $50 on notebooks that cost $6 each. How many notebooks can the student buy?**

Let _____

Equation: _____

Solve (show each step):

Answer: _____

✏ Practice: Word Problems

1. ★ Try it Tickets cost $7 each. Maya spent $42. How many tickets did she buy?

Answer: _____

2. ★★ On level Which equation matches: “Twice a number, decreased by 5, is 17”?

- ○ $2x - 5 = 17$
- ○ $2(x - 5) = 17$
- ○ $5 - 2x = 17$
- ○ $2x + 5 = 17$

3. ★★ On level A plumber charges $50 plus $30 per hour. The bill was $170. How many hours did the plumber work?

Answer: _____

4. ★★★ Challenge Jordan has $25. There is a $3 fee, and each snack costs $4. What is the greatest number of snacks Jordan can buy?

Answer: _____

✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

This part mixes problems from **all 8 sections**. Every problem has a level: ★ Try it ★★ On level ★★★ Challenge. Stuck? Do the ★ problems first, then look back at that section of the notes.

A. Multi-Step Equations Sections 2–5

1) ★ Try it $3x + 7 = 22$

SOLVE $3x + 7 = 22$

Answer: _____

2) ★ Try it $2(x - 4) = 10$

SOLVE $2(x - 4) = 10$

Answer: _____

3) ★★ On level $5x - 3 = 2x + 9$

SOLVE $5x - 3 = 2x + 9$

Answer: _____

4) ★★ On level $-4(x + 2) = 12$

SOLVE $-4(x + 2) = 12$

Answer: _____

5) ★★ On level $6x + 4 = 4x - 8$

SOLVE $6x + 4 = 4x - 8$

Answer: _____

6) ★★★ Challenge $3(2x + 1) = 4x + 7$

SOLVE $3(2x + 1) = 4x + 7$

Answer: _____

7) ★ Try it $\frac{x}{4} + 5 = 8$

SOLVE $\frac{x}{4} + 5 = 8$

Answer: _____

8) ★★ On level $7 - 2x = 3x + 17$

SOLVE $7 - 2x = 3x + 17$

Answer: _____

9) ★★ On level $0.5x + 3 = 7$

SOLVE $0.5x + 3 = 7$

Answer: _____

10) ★★★ Challenge $-(x - 6) = 2x + 3$

SOLVE $-(x - 6) = 2x + 3$

Answer: _____

B. Special Cases — No Solution or Infinite Solutions? Section 6

1) ★ Try it $2x + 3 = 2x + 3$

SOLVE $2x + 3 = 2x + 3$

Answer: _____

2) ★ Try it $4x + 1 = 4x - 5$

SOLVE $4x + 1 = 4x - 5$

Answer: _____

3) ★★ On level $3(x + 2) = 3x + 6$

SOLVE $3(x + 2) = 3x + 6$

Answer: _____

4) ★★ On level $5x - 5 = 5(x + 1)$

SOLVE $5x - 5 = 5(x + 1)$

Answer: _____

5) ★ Try it $7x + 2 = 7x + 2$

SOLVE $7x + 2 = 7x + 2$

Answer: _____

6) ★ Try it $x + 4 = x - 1$

SOLVE $x + 4 = x - 1$

Answer: _____

C. Inequalities Section 7

Solve it like an equation. FLIP the sign only when you multiply or divide both sides by a negative number.

1) ★ Try it $3x + 2 > 11$

SOLVE $3x + 2 > 11$

Answer: _____

Graph the answer:



2) ★ Try it $2x - 5 \leq 7$

SOLVE $2x - 5 \leq 7$

Answer: _____

Graph the answer:



3) ★★ On level $-4x > 12$ (flip!)

SOLVE $-4x > 12$

Answer: _____

Graph the answer:



4) ★ Try it $5x + 1 \geq 16$

SOLVE $5x + 1 \geq 16$

Answer: _____

Graph the answer:



5) ★★★ Challenge $7 - x < 4$ (flip!)

SOLVE $7 - x < 4$

Answer: _____

Graph the answer:



6) ★★ On level $2(x + 3) \leq 10$

SOLVE $2(x + 3) \leq 10$

Answer: _____

Graph the answer:



7) ★★★ Challenge $-3x + 9 \geq 0$ (flip!)

SOLVE $-3x + 9 \geq 0$

Answer: _____

Graph the answer:



8) ★★ On level $4x - 1 > 2x + 5$

SOLVE $4x - 1 > 2x + 5$

Answer: _____

Graph the answer:



D. Word Problems Section 8

1. ★★ On level A phone plan costs $35 per month plus $0.10 per text. Last month's bill was $52. How many texts were sent?

Let _____

Equation: _____

Solve (show each step):

Answer: _____

2. ★★★ Challenge Sam has $240 and saves $30 per week. Jake has $420 and spends $10 per week. When will they have the same amount?

Let _____

Equation: _____

Solve (show each step):

Answer: _____

3. ★★★ Challenge A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

Let _____

Equation: _____

Solve (show each step):

Answer: _____

4. ★★ On level Movie tickets cost $9 each. You have at most $50 to spend. What is the maximum number of tickets?

Let _____

Equation: _____

Solve (show each step):

Answer: _____

5. ★★★ Challenge A car rental costs $45 per day plus $0.25 per mile. Your budget is $120 for one day. How many miles can you drive?

Let _____

Equation: _____

Solve (show each step):

Answer: _____

6. ★★★ Challenge Ana, Ben and Cal earned $78 in all. Ben earned $2 more than Ana. Cal earned $4 more than Ana. How much did Ana earn?

Let _____

Equation: _____

Solve (show each step):

Answer: _____

E. Mixed Review

Sections 1–8

✏ Comprehensive Practice

1. ★ Try it Section 1 · Solve: $x - 7 = -2$

Answer: _____

2. ★ Try it Section 2 · Solve: 5x − 4 = 26

Answer: _____

3. ★ Try it Section 7 · Which inequality means “x is at most 12”?

- ○ x ≤ 12
- ○ x < 12
- ○ x ≥ 12
- ○ x > 12

4. ★★ On level Section 3 · Solve: −3(x + 4) = 6

Answer: _____

5. ★★ On level Section 4 · Solve: 9x − 2 = 4x + 23

Answer: _____

6. ★★ On level Section 6 · How many solutions does 3(2x − 1) = 6x + 3 have?

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions

7. ★★ On level Section 7 · Solve: $8 - 3x \leq 20$.

Answer: _____

8. ★★★ Challenge Section 5 · Solve: $2(3x - 1) + 4 = 4(x + 2)$

Answer: _____

9. ★★★ Challenge Section 8 · The sum of three times a number and 8 is −13. What is the number?

Answer: _____

10. ★★★ Challenge Section 8 · Ava has 18 stickers and gets 4 more each day. Ben has 50 stickers and gives away 4 each day. After how many days do they have the same number?

Answer: _____